Problem Set 1

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1 The Zero Vector

Let V be a vector space over $\mathbb{C}$ with zero vector $|0\rangle$. Then

$$0|v\rangle \overset{*}{=} (0+0)|v\rangle \overset{4}{=} 0|v\rangle + 0|v\rangle \overset{8,9}{\implies} 0|v\rangle + (0|v\rangle - 0|v\rangle) \overset{8}{=} 0|v\rangle \overset{\star}{=} 0|v\rangle - 0|v\rangle \overset{8}{=} |0\rangle.$$

Note that the step denoted by $*$ follows by the field axioms, rather than the vector space axioms. Additionally, the step denoted $\star$ follows by subtracting $|0\rangle$ from both sides of the previous implication that $0|v\rangle + 0|v\rangle = 0|v\rangle$; I just moved stuff around to make it fit on one line.

2 Vector Spaces

For all of these problems, let V be the given set, and define addition and scalar multiplication pointwise; i.e. $(f+g)(x) = f(x) + g(x)$ and $(zf)(x) = z \cdot f(x)$ for all $f, g \in V$ and $z \in \mathbb{C}$.

1. Yes.
   (a) Closure under addition: let $f, g \in V$. We have

   $$(f+g)(0) = f(0) + g(0) = 0 + 0 = 0 \text{ and } (f+g)(L) = f(L) + g(L) = 0 + 0 = 0.$$

   Therefore, $f + g \in V$;

   (b) Closure under scalar multiplication: let $z \in \mathbb{C}$. We have

   $$(zf)(0) = z(f(0)) = z0 = 0 \text{ and } (zf)(L) = z(f(L)) = z0 = 0,$$

   so $zf \in V$.

2. No. Using the same definitions as before, note that for any $f \in V$, $f'(0) > 0$. Note that $-1 \in \mathbb{C}$. We have

   $$\frac{\mathrm{d}}{\mathrm{d}x}(-1f)(x) = -f'(x).$$

   It follows that $(-1f)'(0) = -f'(0) < 0$, so $-1f \notin V$.

3. Yes. It suffices to show the only element of the space is the 0 function, because $0 + 0 = 0$ and $\alpha 0 = 0$ for all 0 vectors in all vector spaces, and all scalars α in all fields. Let $f \in V$. Since f has a vanishing first derivative everywhere, it follows that f is continuous everywhere. Let $x \in [0, L]$ and suppose for contradiction that $f(x) > 0$. Since f is continuous on $[0, L]$, by the intermediate value theorem, f takes on every value on the interval $[0, f(x)]$ at some point on the interval $[0, x]$. Since f is continuous and $f(x) \gneq 0$, we have that f is increasing on some subinterval of $[0, f(x)]$; equivalently, f has a positive first derivative on this interval. Therefore, $f(x) = 0$ for all $x \in [0, L]$. Note that this implies $f(L/2) = 0$, so that property is also satisfied. Therefore, $f \in V$ implies $f = 0$, so $V = \{0\}$ is the trivial vector space.

3 Eigenvalues

1. Notice that $M = |u\rangle\langle u|$. Let $\lambda|x\rangle = M|x\rangle$. We have

$$\lambda|x\rangle = |u\rangle\langle u\,|\,x\rangle = |u\rangle\sum_{n=1}^{10} nx_n.$$

   We will use index notation to denote this as $|u\rangle nx_n$. We have

$$\lambda x_k = u_k nx_n = knx_n.$$

   We have

$$\lambda x_1 = nx_n.$$

   Additionally,

$$\lambda x_k = knx_n = k(\lambda x_1) \implies x_k = kx_1.$$

   It follows that

$$\lambda x_1 = n(nx_1) = x_1\sum_{n=1}^{10} n^2.$$

   Therefore,

$$\lambda = \sum_{n=1}^{10} n^2 = 385.$$

   Moreover, note that the rank of the matrix is 1:

```python
#!/bin/python3

import numpy as np

u = np.arange(1, 11)
u = u.reshape(-1, 1)

print(np.linalg.matrix_rank(u @ u.T))
```

   This program returns 1. The number of nonzero eigenvalues of a symmetric matrix is equal to its rank, and we implicitly assumed $\lambda \neq 0$ when we stated $x_k = kx_1$. If we suppose $\lambda = 0$, then we have $nx_n = 0$, since $k \neq 0$ for all k. We can define $x_k = (-1)^k 1/k$, and note that

$$\sum_{n=1}^{10}(-1)^n\frac{n}{n} = 5 - 5 = 0.$$

   Therefore, 0 is also an eigenvalue for at least this singular vector.

2. I got a fever so I'm not going to do this the creative way and am just going to write a script.

```python
#!/bin/python3

import numpy as np

u = np.arange(1, 11)
u = u.reshape(-1, 1)

v = np.arange(1, 11) / 2
```

```python
v = v.reshape(-1, 1)

w = np.arange(1, 11)
w = (-1) ** w / w
w = w.reshape(-1, 1)

print(np.linalg.eigvals((u @ u.T) + (v @ v.T) + (w @ w.T)))

```

This outputs

```
    [ 4.81250000e+02+0.00000000e+00j  1.54976773e+00+0.00000000e+00j
  7.78879923e-15+0.00000000e+00j -9.48602991e-15+3.10320984e-15j
 -9.48602991e-15-3.10320984e-15j  1.96268824e-15+1.23172745e-15j
  1.96268824e-15-1.23172745e-15j -2.11183098e-15+1.96936937e-15j
 -2.11183098e-15-1.96936937e-15j -6.23523255e-16+0.00000000e+00j]

```

Since this matrix is the sum of three rank 1 matrices, it's rank should be no more than 3, so it should have no more than 3 nonzero eigenvalues. This implies that the last 8 eigenvalues are floating point errors, since they are very small. Moreover, running the same code but changing the last line to print `np.linalg.matrix_rank(...)`, we find that the rank of the matrix is precisely 2. Therefore, the only actual eigenvalues are

$$481.25, \qquad 1.55.$$